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Paper Code : BS-M401/M(CS) 401/M(CS)401 Numerical Methods(BS)

UPID : 004401

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is the limitation of performing numerical integration using Simpson's 1/3 Rule?

(II) To eliminate steps in Gauss elimination method to reduce the coefficient matrix, which part of matrix is used?

(III) The first approximation of real root of $x^3 - 2x^2 - 2 = 0$ by Regula Falsi Method between 2 and 3 is_____

(IV) The velocity (m/s) of a body is given as a function of time (seconds) by v(t)=200 ln(1+t) -t, t≥0
Using Euler's method with a step size of 5 seconds, calculate the distance in meters travelled by the body from t=2 to t=12

(V) Round off 38.5218 to one significant figure

(VI) If f(x)=1/x, then what is f(a,b,c) using Divided difference?

(VII) To evaluate a definite integral of f(x) within lower limit 'a' and upper limit 'b' using Simpson's 1/3 rule, [a, b] should be divided into _____ subintervals.

(VIII) Find the solution of differential equation $\frac{dy}{dx} = 1 + \frac{y}{x}, y(1) = 1.$

(IX) What is the iterative formula for Newton Raphson method?

(X) Find the value of y at x = 2 for the given set of points (1, 2),(3, 4) using Lagrange Interpolation.

(XI) When do you encounter truncation error?

(XII) Fill in the blanks:
The _____and _____interpolation methods result in the smallest polynomia function, i.e., the polynomial of the lowest possible degree that goes through the data points in the data se

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. State one advantage and disadvantage of fixed point number representation. [5]

3. A second degree polynomial passes through the points (1,-1), (2,-1),(3,1),(4,5). Find the polynomial. [5]

4. Integrate f(x)= 1/x from =1 to 2 by Simpson's one third Rule (taking four subintervals) correct to 5-decimal places. [5]

5. Solve the following system of linear equations using Gauss Elimination method. [5]
$2x_1+8x_2+2x_3 = 14$
$x_1+6x_2-x_3 = 13$
$2x_1-x_2+2x_3 = 5$

6. Find the smallest positive roots of $3x^3-9x^2+8=0$ and Newton Raphson method correct up to 5 decimal places. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Find a real root of the equation $f(x) = x^3-9x+1 = 0$ lying between 2 and 3 correct to 3 significant figure. [7]

(b) Explain how roots of a function can be evaluated by Bisection Method. [8]

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8. (a) How can LU decomposition method be used to solve a set of linear equation? [4]

(b) Use LU decomposition to solve for x and y in the given equations : [7]
2x+y=7 and x+2y=5

(c) What is a diagonally dominant matrix? Name and explain one method which uses diagonally dominant matrix to solve a set of given linear equations. [4]

9. (a) Use Newton's Divided difference to find f(6) for given set of data points (x,f(x)) which are (5, 150), (7, 392),(11, 1452),(13, 2366), (21, 9702). [8]

(b) Can the above question be done by Lagrange Interpolation. State advantage and disadvantage of Lagrange compared to Newton's divided difference. [7]

10. (a) Find values of y(0.2) and y(0.4) using Runge Kutta 4th Order Method with step size 0.2 for ODE $\frac{dy}{dx} = \frac{1}{2}(x^2 + y)$, given y(0)=0.8. [10]

(b) Compare Runge Kutta 4th order method and Milne's Predictor Corrector Method for solving ODE. [5]

11. (a) Use Newton-Raphson to deduce the iterative procedure $x_{n+1} = \frac{1}{2}\left(x_n + \frac{y}{x_n}\right)$ for evaluating y(0.5) as the solution of equation x^2-y = 0. [8]

(b) Calculate the positive root of the equation x^2+2x-2 = 0. Correct up to two significant figures by the Newton-Raphson method. [7]

*** END OF PAPER ***

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